

Generation Interconnection System Impact Study Report

For Queue Project Q18-0916

Revere 100 kV

March 17, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0916. The Interconnection Customer proposes a 178.3 MW solar photovoltaic facility interconnecting at the Revere 100 kV substation in the Indiana area. The request was received on January 11, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 367 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,843,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Hebo-Barrett 230 kV	Hebo-Winthrop 230 kV line	135.6	560	5.8
Troy Energy Llc-Highgate 230 kV	Trego-Troy Energy Llc 500 kV	120.6	951	11.7
Baraga Vlg 500-230 kV Transformer Bank	Trego Substation-Beaver 500 kV line	118.5	1,240	13.3
St. Lawrence Substation-James De Young 230 kV L	E Methuen-Lonetree 500 kV	117.9	1,196	7.2
Galion Generating Station 500 kV/230 kV XFMR	Beaver-Trego 500 kV L	115.9	1,344	10.7
Eufaula 230-1 kV Transformer Bank	Imperial-Eufaula 500 kV	115.5	1,922	12.2
Eufaula 500/1 kV Autotransformer	Imperial-Eufaula 500 kV	115.5	1,922	12.2
Post Rock Substation-Newport Tie 230 kV	Beaver-Trego 500 kV L	115.4	1,766	13.2
Beaver 500 kV/1 kV Auto	Eufaula-Beaver 500 kV	114.6	2,321	13.7
Beaver 230 kV/1 kV Autotransformer	Beaver-Eufaula 500 kV	114.6	2,321	13.7
Post Rock-Myrtle Pt 230 kV	Trego-Beavr 500 kV line	114.3	1,820	12.7
Swan Vly-Eastown Rd 230 kV	Swan Valley-Trego 500 kV line	113.8	784	6.1
Onancock Diesel Generation Facility-Beaver 230 kV	Beaver-Eufaula 500 kV	113.5	1,709	10.8
Highgate-Herald-Twin Cities 230 kV	Masn St 500 kV/230 kV Auto	112.1	1,098	6.9
Eufaula-Imperial 500 kV	Galion Generating Sta-Trego 500 kV line	111.4	3,337	28.9
Eufaula-Beaver 500 kV	Trego-Galion Generating Station 500 kV	111.3	3,975	29.6
Herald-Twin Cities-Mason St 230 kV	Mason St-New Departure 230 kV	110.9	734	5.3

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Gruver-Galeville 230 kV L	Eufaula-Imperial 500 kV L	110.6	622	6.5
Trego 500/1 kV Autotransformer	Trego-Troy Energy Llc 500 kV	110.4	769	12.4
Adams-New Departure 230 kV Line	Trego-Beaver Substation 500 kV Line	110.3	1,129	7.7
Canal Road-Newport Tie 230 kV	Beaver-Trego 500 kV	109.9	1,173	10.4
Mason St Substation 500 kV/230 kV TX	Trego-Beaver 500 kV L	109.7	1,705	11.7
Trego-Troy Energy Llc 500 kV	Swan Vly-Trego 500 kV Line	109.5	3,272	31.1
E Methuen-Lonetree 500 kV	Muskogee-Montpelier Electric Generating Station 500 kV L	107.8	3,188	10.3
Sabanera-Lonetree 500 kV	Muskogee-Montpelier Electric Generating Station 500 kV L	107.8	3,482	11.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Barrett-Hebo 230 kV	Rebuild line with higher-capacity conductor	\$10,302,000	1% (\$102,000)
Highgate-Troy Energy Llc 230 kV line	Rebuild line with higher-capacity conductor	\$71,674,000	1% (\$1,005,000)
Baraga Vlg 500-230 kV Transformer Bank	Replace transformer with higher-rated unit	\$21,229,000	1% (\$215,000)
St. Lawrence Substation-James De Young 230 kV L	Rebuild line with higher-capacity conductor	\$73,566,000	1% (\$650,000)
Galion Generating Sta Substation 500/230kV XFMR	Replace transformer with higher-rated unit	\$35,015,000	1% (\$391,000)
Eufaula 230-1 kV Transformer Bank	Replace transformer with higher-rated unit	\$7,513,000	1% (\$82,000)
Eufaula 500 kV/1 kV TX	Replace transformer with higher-rated unit	\$30,322,000	1% (\$331,000)
Post Rock Substation-Newport Tie 230 kV	Rebuild line with higher-capacity conductor	\$59,043,000	1% (\$640,000)
Beaver 500-1 kV Transformer	Replace transformer with higher-rated unit	\$26,148,000	1% (\$284,000)
Beaver 230 kV/1 kV Autotransformer	Replace transformer with higher-rated unit	\$5,196,000	1% (\$56,000)
Post Rock-Myrtle Pt 230 kV	Rebuild line with higher-capacity conductor	\$23,610,000	1% (\$256,000)
Swan Valley-Eastown Road 230 kV	Rebuild line with higher-capacity conductor	\$14,261,000	1% (\$199,000)
Beaver-Onancock Diesel Generation Facility 230 kV	Rebuild line with higher-capacity conductor	\$20,037,000	1% (\$217,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Herald-Twin Cities-Highgate 230 kV L	Rebuild line with higher-capacity conductor	\$25,839,000	1% (\$321,000)
Imperial-Eufaula 500 kV Line	Rebuild line with higher-capacity conductor	\$83,090,000	1% (\$922,000)
Beaver-Eufaula 500 kV	Rebuild line with higher-capacity conductor	\$90,761,000	1% (\$1,003,000)
Mason St-Herald-Twin Cities 230 kV	Rebuild line with higher-capacity conductor	\$34,249,000	2% (\$544,000)
Gruver-Galeville 230 kV L	Rebuild line with higher-capacity conductor	\$14,762,000	1% (\$164,000)
Trego 500/1 kV Autotransformer	Replace transformer with higher-rated unit	\$17,991,000	1% (\$251,000)
Adams-New Departure 230 kV Line	Rebuild line with higher-capacity conductor	\$26,906,000	1% (\$306,000)
Canal Road-Newport Tie 230 kV	Reconductor line	\$5,073,000	1% (\$55,000)
Mason St Substation 500 kV/230 kV TX	Replace transformer with higher-rated unit	\$27,650,000	1% (\$321,000)
Trego-Troy Energy Llc 500 kV	Reconductor line	\$62,101,000	1% (\$767,000)
East Methuen-Lonetree 500 kV	Reconductor line	\$45,781,000	1% (\$312,000)
Sabanera-Lonetree 500 kV	Reconductor line	\$63,400,000	1% (\$449,000)
	Total allocated to project		\$9,843,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.